ChatGPT said:

For an autonomous vehicle, **pose estimation** means continuously estimating the vehicle's 3D position and orientation—and usually velocity and IMU biases—from IMU, GNSS, wheel odometry, LiDAR and/or cameras.

The central quantity is

$$\boxed{T_{WE}(t) = \begin{bmatrix} R_{WE}(t) & p_{WE}(t) \\ 0 & 1 \end{bmatrix} \in SE(3)}$$

where R_{WE} is orientation and p_{WE} is position of the ego vehicle in the world frame.

A production estimator is essentially:

$$\boxed{\text{IMU propagation} + \text{GNSS/LiDAR/Camera/Wheel corrections} + \text{uncertainty propaga}}$$

Below is the mathematical treatment from rigid-body pose through IMU mechanization, error-state EKF, sensor measurement models, and optimization.

---

1. What exactly are we estimating?

A minimal pose consists of

$$p = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \qquad R \in SO(3).$$

But an AV localization system generally estimates more:

$$\boxed{x = [p,\ v,\ R,\ b_a,\ b_g]}$$

where

$$p \in \mathbb{R}^3$$

is position,

$$v \in \mathbb{R}^3$$

is velocity,

$$R \in SO(3)$$

is orientation,

$$b_a \in \mathbb{R}^3$$

is accelerometer bias, and

$$b_g \in \mathbb{R}^3$$

is gyroscope bias.

So the navigation state has nominally

$$3 + 3 + 3 + 3 + 3 = 15$$

degrees of freedom.

Often we also estimate gravity, sensor extrinsics, time offsets, scale factors, etc.

---

2. Coordinate frames

Let's define:

$$\mathcal{W}$$

world/map frame,

$$\mathcal{E}$$

ego/body/IMU frame,

$$\mathcal{L}$$

LiDAR frame,

$$\mathcal{C}$$

camera frame,

and

$$\mathcal{G}$$

GNSS antenna frame if necessary.

Our main pose is

$$T_{WE}.$$

It maps an ego-frame point into world coordinates:

$$\boxed{{}^{W}p = T_{WE}\,{}^{E}p.}$$

---

3. SE(3) representation

Rigid-body pose is

$$T_{WE} = \begin{bmatrix} R_{WE} & p_{WE} \\ 0 & 1 \end{bmatrix}.$$

Its inverse is

$$\boxed{T_{EW} = T_{WE}^{-1} = \begin{bmatrix} R_{WE}^{T} & -R_{WE}^{T}p_{WE} \\ 0 & 1 \end{bmatrix}.}$$

This transformation appears everywhere in LiDAR deskewing, camera projection, tracking and mapping.

---

4. Rotation belongs to $SO(3)$

A valid rotation matrix satisfies

$$R^{T}R = I$$

and

$$\det(R) = 1.$$

Therefore

$$\boxed{R \in SO(3).}$$

We shouldn't treat its nine elements as nine independent state variables.

Common representations are:

- rotation matrix;
- Euler angles;
- quaternion;
- rotation vector / Lie algebra.

For filtering, quaternions plus a 3D small-angle error are common.

---

5. Skew-symmetric matrix

For

$$\omega = \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix},$$

define

$$[\omega]_\times = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}.$$

It has the property

$$[\omega]_\times v = \omega \times v.$$

This is fundamental to rigid-body kinematics.

---

6. Rotation exponential map

A rotation vector

$$\phi = \theta u$$

can be converted into a rotation matrix:

$$R = \mathrm{Exp}([\phi]_\times).$$

Using Rodrigues' formula:

$$\boxed{R = I + \frac{\sin\theta}{\theta}[\phi]_\times + \frac{1-\cos\theta}{\theta^2}[\phi]_\times^2}$$

where

$$\theta = \|\phi\|.$$

For small angles:

$$\boxed{\mathrm{Exp}([\phi]_\times) \approx I + [\phi]_\times.}$$

This becomes extremely useful in the error-state EKF.

---

7. IMU measurements

An IMU gives two primary measurements.

Gyroscope:

$$\boxed{\omega_m = \omega + b_g + n_g}$$

Accelerometer:

$$\boxed{a_m = R_{EW}(a^W - g^W) + b_a + n_a}$$

where:

- b_g: gyro bias;
- b_a: accelerometer bias;
- n_g, n_a: sensor noise;
- g^W: gravity.

Therefore corrected angular velocity is

$$\hat{\omega} = \omega_m - \hat{b}_g.$$

Corrected specific force is

$$\hat{f} = a_m - \hat{b}_a.$$

---

8. Continuous-time navigation equations

The core inertial navigation equations are

$$\boxed{\dot{p} = v}$$

$$\boxed{\dot{v} = R(a_m - b_a - n_a) + g}$$

and

$$\boxed{\dot{R} = R[\omega_m - b_g - n_g]_\times.}$$

Biases are commonly modeled as random walks:

$$\boxed{\dot{b}_a = n_{ba}}$$

$$\boxed{\dot{b}_g = n_{bg}.}$$

These five equations form the basic IMU propagation model.

---

9. Discrete-time IMU propagation

Suppose IMU frequency is 200 Hz:

$$\Delta t = 0.005s.$$

Correct the measurements:

$$\hat{\omega}_k = \omega_{m,k} - \hat{b}_{g,k}$$

and

$$\hat{a}_k = a_{m,k} - \hat{b}_{a,k}.$$

Update orientation:

$$\boxed{R_{k+1} = R_k \,\mathrm{Exp}([\hat{\omega}_k \Delta t]_\times)}$$

Acceleration in world coordinates:

$$\boxed{a_k^W = R_k \hat{a}_k + g^W.}$$

Velocity:

$$\boxed{v_{k+1} = v_k + a_k^W \Delta t.}$$

Position:

$$\boxed{p_{k+1} = p_k + v_k \Delta t + \frac{1}{2} a_k^W \Delta t^2.}$$

This gives a high-rate pose estimate.

---

10. Why IMU alone isn't sufficient

Suppose accelerometer bias is only

$$b_a = 0.05 m/s^2.$$

Ignoring it causes position error approximately

$$e_p(t) = \frac{1}{2} b_a t^2.$$

After 10 seconds:

$$e_p = \frac{1}{2}(0.05)(10^2) = 2.5m.$$

After 60 seconds:

$$e_p = \frac{1}{2}(0.05)(60^2) = 90m.$$

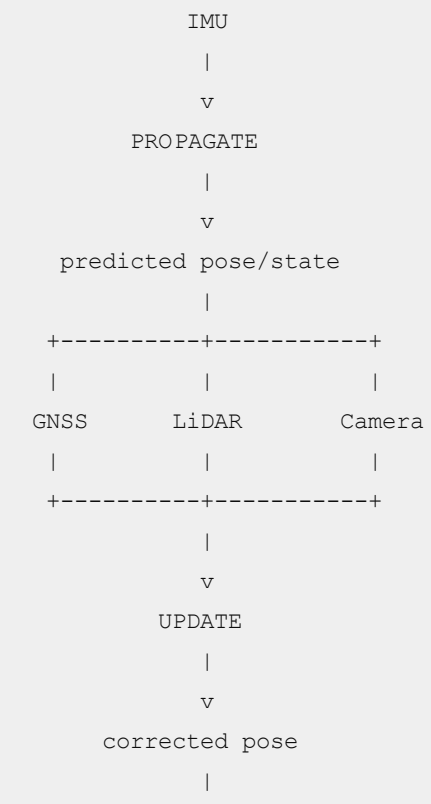
Orientation errors are even more damaging because they cause gravity to project incorrectly into horizontal acceleration.

Therefore IMU gives excellent short-term propagation but requires external corrections.

11. Prediction-correction architecture

The estimator therefore operates as

Plain text



Mathematically:

$$p(x_k|z_{1:k-1})$$

is the predicted state distribution.

New measurement z_k gives

$$\boxed{p(x_k|z_{1:k}) \propto p(z_k|x_k)p(x_k|z_{1:k-1}).}$$

12. Error-State EKF

For AV localization, a particularly useful formulation is the **Error-State Extended Kalman Filter**, ESKF.

Instead of putting orientation directly into an ordinary additive EKF state, maintain:

Nominal state

$$\hat{x} = [\hat{p}, \hat{v}, \hat{R}, \hat{b}_a, \hat{b}_g].$$

Error state

$$\boxed{\delta x = \begin{bmatrix} \delta p \\ \delta v \\ \delta \theta \\ \delta b_a \\ \delta b_g \end{bmatrix} \in \mathbb{R}^{15}.}$$

Covariance is

$$P = E[\delta x \delta x^T].$$

---

13. Orientation error

Instead of

$$R = R + \delta R,$$

which doesn't preserve $SO(3)$, define a multiplicative perturbation.

For one common convention:

$$\boxed{R_{\text{true}} = \hat{R}\,\text{Exp}([\delta\theta]_\times).}$$

For small error,

$$R_{\text{true}} \approx \hat{R}(I + [\delta\theta]_\times).$$

So orientation uncertainty needs only three variables:

$$\delta\theta = \begin{bmatrix} \delta\theta_x \\ \delta\theta_y \\ \delta\theta_z \end{bmatrix}.$$

14. Linearized error dynamics

The ESKF error dynamics can be written

$$\boxed{\delta \dot{x} = F\delta x + Gn.}$$

For

$$\delta x = [\delta p, \delta v, \delta \theta, \delta b_a, \delta b_g]^T,$$

a commonly used first-order structure is

$$F = \begin{bmatrix} 0 & I & 0 & 0 & 0 \\ 0 & 0 & -R[\hat{a}]_\times & -R & 0 \\ 0 & 0 & -[\hat{\omega}]_\times & 0 & -I \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Each entry here represents a 3×3 block.

This matrix contains important physical relationships.

---

15. Interpretation of F

The first row

$$\delta \dot{p} = \delta v$$

says velocity error causes position error.

The second:

$$\boxed{\delta \dot{v} = -R[\hat{a}]_\times \delta\theta - R\delta b_a - Rn_a}$$

shows that acceleration errors arise from:

- orientation error;
- accelerometer bias;
- accelerometer noise.

The third:

$$\boxed{\delta\dot{\theta} = -[\hat{\omega}]_\times \delta\theta - \delta b_g - n_g}$$

shows that orientation error depends strongly on gyro bias.

---

16. Why attitude error corrupts position

This is extremely important.

Suppose the car is stationary.

The accelerometer senses approximately gravity magnitude:

$$g = 9.81 m/s^2.$$

If roll/pitch error is only

$$1^\circ = 0.01745 rad,$$

the apparent horizontal acceleration error can be approximately

$$9.81 \sin(1^\circ) \approx 0.171 m/s^2.$$

After 10 seconds, the corresponding position error can approach

$$\frac{1}{2}(0.171)(10^2) \approx 8.55 m.$$

So accurate attitude estimation is critical even when your primary interest is x, y localization.

---

17. Covariance propagation

Continuous covariance dynamics are

$$\boxed{\dot{P} = FP + PF^T + GQ_cG^T.}$$

For discrete time,

$$\Phi = e^{F\Delta t}.$$

For sufficiently small Δt,

$$\Phi \approx I + F\Delta t.$$

Then:

$$\boxed{P_{k+1} = \Phi P_k \Phi^T + Q_d.}$$

This tells the estimator how uncertainty grows while integrating the IMU.

---

18. GNSS position measurement

Suppose GNSS gives

$$z_G = \begin{bmatrix} x_G \\ y_G \\ z_G \end{bmatrix}.$$

Ignoring antenna lever arm initially,

$$h_G(x) = p.$$

Residual:

$$\boxed{r_G = z_G - \hat{p}.}$$

The Jacobian with respect to the 15D error state is

$$\boxed{H_G = \begin{bmatrix} I & 0 & 0 & 0 & 0 \end{bmatrix}.}$$

Innovation covariance:

$$S = H_G P H_G^T + R_G.$$

Kalman gain:

$$\boxed{K = P H_G^T S^{-1}.}$$

Error correction:

$$\boxed{\delta x = K r_G.}$$

19. Injecting the ESKF correction

Suppose

$$\delta x = [\delta p, \delta v, \delta\theta, \delta b_a, \delta b_g].$$

Update nominal position:

$$\hat{p} \leftarrow \hat{p} + \delta p.$$

Velocity:

$$\hat{v} \leftarrow \hat{v} + \delta v.$$

Biases:

$$\hat{b}_a \leftarrow \hat{b}_a + \delta b_a$$

$$\hat{b}_g \leftarrow \hat{b}_g + \delta b_g.$$

Orientation is multiplicative:

$$\boxed{\hat{R} \leftarrow \hat{R}\,\mathrm{Exp}([\delta\theta]_\times).}$$

Then reset the error state around zero.

---

20. GNSS antenna lever arm

In reality the GNSS antenna is not necessarily at the IMU origin.

Suppose

$$p_{EG}$$

is the antenna position in ego coordinates.

Then predicted GNSS antenna location is

$$\boxed{h_G(x) = p_{WE} + R_{WE} p_{EG}.}$$

Now GNSS measurement depends on both position and orientation.

The residual is

$$r_G = z_G - (p + Rp_{EG}).$$

The attitude Jacobian contains approximately

$$-R[p_{EG}]_\times$$

depending on the perturbation convention.

This is why correct extrinsic calibration matters even for GNSS.

---

21. Wheel odometry

Wheel speed can provide longitudinal velocity.

Suppose the body-frame velocity is

$$v^E = R_{EW} v^W.$$

A wheel-speed measurement can be modeled as

$$\boxed{z_{\text{wheel}} = e_x^T R_{EW} v^W + n_w}$$

where

$$e_x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Wheel odometry is useful because IMU acceleration integration drifts quickly.

---

22. Nonholonomic vehicle constraints

For normal road driving, lateral and vertical body-frame velocities are often small:

$$v_y^E \approx 0$$

$$v_z^E \approx 0.$$

These can be treated as pseudo-measurements:

$$z_{NHC} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

with

$$h(x) = \begin{bmatrix} e_y^T R^T v \\ e_z^T R^T v \end{bmatrix}.$$

This can substantially stabilize ground-vehicle localization.

But it should be weakened during:

- wheel slip;
- aggressive maneuvers;
- rough terrain.

---

23. LiDAR-based pose estimation

LiDAR can estimate ego pose by aligning the current cloud with a map or previous scan.

Suppose current point is

$$p_i^L.$$

With LiDAR extrinsic

$$T_{EL},$$

the point predicted in the world is

$$\boxed{p_i^W = T_{WE} T_{EL} p_i^L.}$$

We seek the pose T_{WE} that best aligns these points with known map geometry.

---

24. Point-to-point ICP

Suppose p_i is a current point and q_i its map correspondence.

Minimize

$$\boxed{J(T) = \sum_i \|q_i - Tp_i\|^2.}$$

More explicitly:

$$J(R,t) = \sum_i \|q_i - (Rp_i + t)\|^2.$$

This is point-to-point ICP.

---

25. Point-to-plane ICP

For autonomous-driving LiDAR, point-to-plane residuals are often more useful.

Suppose map point q_i has surface normal n_i.

Residual:

$$\boxed{r_i = n_i^T(Rp_i + t - q_i).}$$

Optimize:

$$\boxed{\min_{R,t} \sum_i r_i^2.}$$

Usually with robust loss:

$$\min_T \sum_i \rho(r_i^2).$$

---

26. Linearizing LiDAR registration

Perturb pose with

$$T' = T\,\mathrm{Exp}(\delta\xi^\wedge)$$

where

$$\delta\xi = \begin{bmatrix} \delta\rho \\ \delta\phi \end{bmatrix} \in \mathbb{R}^6.$$

For small perturbations,

$$r_i(T') \approx r_i(T) + J_i \delta\xi.$$

Stacking residuals:

$$r = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \end{bmatrix}$$

and Jacobians:

$$J = \begin{bmatrix} J_1 \\ J_2 \\ \vdots \end{bmatrix}.$$

Gauss-Newton solves

$$\boxed{J^T J \delta\xi = -J^T r.}$$

Therefore

$$\boxed{\delta\xi = -(J^T J)^{-1} J^T r.}$$

Update:

$$\boxed{T \leftarrow T \,\mathrm{Exp}(\delta\xi^\wedge).}$$

Repeat until convergence.

---

27. Camera pose estimation

A camera estimates pose using image observations of known or reconstructed 3D points.

Suppose a world landmark is

$$P_i^W.$$

Transform into camera:

$$P_i^C = T_{CW} P_i^W.$$

Let

$$P_i^C = \begin{bmatrix} X_i \\ Y_i \\ Z_i \end{bmatrix}.$$

Projection:

$$\pi(P_i^C) = \begin{bmatrix} f_x X_i / Z_i + c_x \\ f_y Y_i / Z_i + c_y \end{bmatrix}.$$

Observed image feature:

$$z_i = \begin{bmatrix} u_i \\ v_i \end{bmatrix}.$$

Reprojection residual:

$$\boxed{r_i = z_i - \pi(T_{CW} P_i^W).}$$

Then optimize:

$$\boxed{T^* = \arg\min_T \sum_i \| z_i - \pi(T P_i) \|^2.}$$

This is the basis of PnP, visual odometry and bundle adjustment.

---

28. Camera projection Jacobian

For

$$P_C = [X, Y, Z]^T,$$

projection is

$$u = f_x X/Z + c_x$$

$$v = f_y Y/Z + c_y.$$

The projection Jacobian is

$$\boxed{J_\pi = \begin{bmatrix} f_x/Z & 0 & -f_x X/Z^2 \\ 0 & f_y/Z & -f_y Y/Z^2 \end{bmatrix}.}$$

Combine this with the SE(3) pose Jacobian using the chain rule to optimize camera pose.

---

29. Visual odometry

Suppose camera poses are

$$T_0, T_1, T_2, \ldots$$

Image features provide relative transformations:

$$T_{C_{k+1}C_k}.$$

Then

$$T_{WC_{k+1}} = T_{WC_k} T_{C_kC_{k+1}}$$

subject to transformation convention.

The important concept is that VO provides **relative motion**, while GNSS/map localization can provide absolute correction.

---

30. LiDAR odometry

Similarly:

$$\mathcal{P}_k \to \mathcal{P}_{k+1}$$

through scan matching estimates

$$T_{k+1 \leftarrow k}.$$

Then pose is accumulated:

$$T_{W,k+1} = T_{W,k} T_{k \to k+1}.$$

But accumulating relative transforms causes drift.

Therefore map matching, GNSS or loop closure is needed.

---

31. IMU + LiDAR fusion

A powerful architecture is:

$$\boxed{\text{IMU predicts pose} \rightarrow \text{LiDAR registration corrects pose.}}$$

At high rate:

$$\hat{x}_{k|k-1} = f(\hat{x}_{k-1}, u_k).$$

LiDAR scan matching provides

$$T_{\text{LiDAR}}.$$

Form a pose residual on $SE(3)$:

$$\boxed{r_L = \text{Log}\left(T_{\text{pred}}^{-1} T_{\text{LiDAR}}\right).}$$

This residual is a 6D vector:

$$r_L = \begin{bmatrix} r_{\text{translation}} \\ r_{\text{rotation}} \end{bmatrix}.$$

It can then update the filter.

---

32. IMU + camera fusion

Similarly:

```
Plain text

IMU
 |
```

```
 | 200-1000 Hz
 v
pose propagation
 |
 +---------------- Camera
 |                    |
 |              feature tracking
 |                    |
 |              reprojection
 |                 residual
 |                    |
 +--------- update ---+
```

The camera provides strong relative geometric constraints while IMU gives high-frequency dynamics and metric inertial information.

This is the foundation of visual-inertial odometry.

---

33. IMU preintegration

Optimization-based estimators often don't integrate every IMU measurement directly inside every optimization iteration.

Between keyframes i and j, preintegrate IMU measurements into:

$$\boxed{\Delta R_{ij}, \quad \Delta v_{ij}, \quad \Delta p_{ij}.}$$

Approximately,

$$\Delta R_{ij} = \prod_{k=i}^{j-1} \mathrm{Exp}[(\omega_k - b_g)\Delta t]_\times.$$

Velocity increment:

$$\Delta v_{ij} \approx \sum_k \Delta R_{ik}(a_k - b_a)\Delta t.$$

Position increment:

$$\Delta p_{ij} \approx \sum_k \left[\Delta v_{ik}\Delta t + \frac{1}{2}\Delta R_{ik}(a_k - b_a)\Delta t^2\right].$$

These form an IMU factor between states.

34. Factor-graph formulation

Instead of an EKF, modern localization systems can solve a nonlinear optimization problem.

At each keyframe:

$$X_k = \{T_k, v_k, b_{a,k}, b_{g,k}\}.$$

Factors can include:

Plain text

```
        GNSS
          |
          v
X0 -- IMU -- X1 -- IMU -- X2 -- IMU -- X3
|             |             |             |
LiDAR       Camera        LiDAR         GNSS
|             |             |             |
Map           Map           Map           Map
```

We seek

$$\boxed{X^* = \arg\min_X \sum_i \|r_i(X)\|^2_{\Sigma_i^{-1}}.}$$

where

$$\|r\|^2_{\Sigma^{-1}} = r^T \Sigma^{-1} r.$$

35. Maximum likelihood interpretation

If sensor errors are Gaussian,

$$z_i = h_i(x) + n_i$$

with

$$n_i \sim \mathcal{N}(0, R_i),$$

then

$$p(z_i|x) \propto \exp\left(-\frac{1}{2} r_i^T R_i^{-1} r_i\right).$$

Maximizing likelihood is therefore equivalent to minimizing:

$$\boxed{\sum_i r_i^T R_i^{-1} r_i.}$$

This explains why covariance-weighted least squares appears everywhere in pose estimation.

---

36. MAP estimation

Include a prior:

$$p(X|Z) \propto p(Z|X)p(X).$$

Then:

$$\boxed{X^* = \arg\min_X \left[\|r_{\text{prior}}\|^2_{\Sigma_{\text{prior}}^{-1}} + \sum_i \|r_i(X)\|^2_{\Sigma_i^{-1}}\right].}$$

This is maximum a posteriori estimation.

---

37. Gauss-Newton for pose estimation

Suppose the total residual is

$$r(x).$$

Linearize around current estimate:

$$r(x + \delta x) \approx r(x) + J\delta x.$$

We want

$$\min_{\delta x} \|r + J\delta x\|^2_W.$$

Setting derivative to zero gives

$$\boxed{J^T W J\, \delta x = -J^T W r.}$$

Define

$$H = J^T W J$$

and

$$b = J^T W r.$$

Then:

$$\boxed{H\delta x = -b.}$$

Solve, update the state, relinearize and repeat.

---

38. EKF and nonlinear least squares are closely related

EKF measurement update:

$$K = PH^T(HPH^T + R)^{-1}.$$

Optimization:

$$(J^T R^{-1} J + P^{-1})\delta x = J^T R^{-1} r.$$

Both are essentially combining:

$$\boxed{\text{prior information} + \text{measurement information.}}$$

The implementation and linearization strategy differ, but the probabilistic foundation is closely related.

---

39. Pose covariance

The estimator should never output only:

$$x = 123.2m, \quad y = 52.8m, \quad \psi = 37^\circ.$$

It should also estimate uncertainty.

For a 6D pose error:

$$\delta\xi = [\delta x, \delta y, \delta z, \delta\theta_x, \delta\theta_y, \delta\theta_z]^T,$$

we have

$$\boxed{\Sigma_{\text{pose}} = E[\delta\xi\delta\xi^T].}$$

For example:

$$\sigma_x = 0.05m, \qquad \sigma_y = 0.08m, \qquad \sigma_\psi = 0.2^\circ.$$

This uncertainty is extremely valuable downstream.

---

40. Innovation gating

Suppose GNSS suddenly reports a position 20 m away because of multipath.

Residual:

$$r = z - h(\hat{x}).$$

Innovation covariance:

$$S = HPH^T + R.$$

Mahalanobis distance:

$$\boxed{d_M^2 = r^T S^{-1} r.}$$

Reject or downweight the measurement if

$$d_M^2 > \gamma.$$

This is the same statistical concept we used in the camera-LiDAR tracker.

---

41. Robust loss

Optimization systems can also use robust losses:

$$\min \sum_i \rho\left(r_i^T \Sigma_i^{-1} r_i\right).$$

Examples include Huber and Cauchy losses.

This reduces the influence of:

- incorrect visual correspondences;
- bad LiDAR matches;
- GNSS multipath;
- dynamic objects.

---

42. Observability

Not every state is always observable.

For example, if the car is stationary, some IMU biases and heading information can be difficult to estimate without additional information.

In a featureless straight tunnel:

- GNSS may disappear;
- camera may have poor features;
- LiDAR geometry may become degenerate.

Suppose LiDAR sees essentially two parallel walls.

Certain directions of motion can have weak geometric constraints.

Mathematically this appears as small eigenvalues of the information/Hessian matrix:

$$H = J^T W J.$$

If

$$\lambda_{\min}(H) \approx 0,$$

there is a poorly observable direction.

---

43. Ground vehicle constraints help observability

Cars have useful physical constraints.

For normal driving:

$$v_z \approx 0$$

and often

$$v_y^{body} \approx 0.$$

Roll and pitch are also often strongly constrained by gravity/road geometry.

These constraints reduce drift.

---

44. Initialization

Pose estimation needs careful initialization.

Orientation

Roll and pitch can often be initialized from stationary accelerometer measurements:

$$a_m \approx -R^T g.$$

Gravity gives two orientation DOF.

But gravity does not determine yaw.

Yaw needs information such as:

- GNSS heading;
- vehicle motion;
- magnetometer where appropriate;
- visual/LiDAR map alignment.

Position

GNSS or map localization can initialize global position.

Velocity

GNSS velocity or wheel speed can initialize velocity.

Biases

When stationary:

$$\omega_m \approx b_g$$

so

$$\hat{b}_g \approx \frac{1}{N}\sum_k \omega_{m,k}.$$

Accelerometer bias can similarly be estimated after accounting for gravity and orientation.

---

45. Extrinsic calibration

Every sensor has its own frame.

For LiDAR:

$$T_{EL}.$$

For camera:

$$T_{EC}.$$

For GNSS:

$$T_{EG}.$$

For example:

$$\boxed{T_{WL} = T_{WE}T_{EL}.}$$

Camera:

$$\boxed{T_{WC} = T_{WE}T_{EC}.}$$

Poor extrinsic calibration directly contaminates pose estimation.

---

46. Temporal calibration

Sensor timestamps also matter.

Suppose camera time offset is

$$\Delta t_C.$$

Then the camera measurement actually corresponds to

$$t_C + \Delta t_C.$$

At $20 m/s$, a 10 ms timing error corresponds to:

$$20(0.01) = 0.2m.$$

So pose estimation and LiDAR deskewing depend on accurate temporal calibration as much as spatial calibration.

---

47. Relationship to LiDAR deskewing

This connects directly to your previous question.

Pose estimator provides

$$\boxed{T_{WE}(t)}$$

at high temporal resolution.

For LiDAR point i, measured at t_i:

$$p_i^W = T_{WE}(t_i) T_{EL} p_i^L.$$

To deskew to reference ego time t_r:

$$\boxed{p_i^{E_r} = T_{WE}(t_r)^{-1} T_{WE}(t_i) T_{EL} p_i^L.}$$

So:

$$\boxed{\text{accurate deskewing fundamentally depends on accurate pose estimation.}}$$

---

48. Relationship to object-motion compensation

For a dynamic object:

$$\boxed{p_i^{E_r} = T_{WE}(t_r)^{-1} T_{WO}(t_r) T_{WO}(t_i)^{-1} T_{WE}(t_i) T_{EL} p_i^L.}$$

Now there are actually **two pose-estimation problems**:

$$T_{WE}(t)$$

ego pose estimation,

and

$$T_{WO_j}(t)$$

tracked object pose estimation.

The ego localization system estimates the first.

The multi-object tracker estimates the second.

---

49. Production AV localization stack

A realistic architecture therefore looks like:

Plain text

```
                        GNSS
                          |
                          v
Camera -------> Visual odometry / landmarks
   |                      |
   |                      v
LiDAR --------> Scan / map matching
   |                      |
   |                      v
Wheel --------> Vehicle odometry
                          |
IMU ----------------------+
 |                        |
 | 200-1000 Hz            |
 v                        v
Inertial propagation ---> Sensor fusion
                          |
                    ESKF / smoother
                          |
                          v
```

---

50. Complete mathematical summary

The nominal state is

$$\boxed{x = [p, v, R, b_a, b_g].}$$

IMU propagation:

$$\boxed{\dot{p} = v}$$

$$\boxed{\dot{v} = R(a_m - b_a) + g}$$

$$\boxed{\dot{R} = R[\omega_m - b_g]_\times}$$

$$\boxed{\dot{b}_a = n_{ba}, \qquad \dot{b}_g = n_{bg}.}$$

Discrete orientation:

$$\boxed{R_{k+1} = R_k \,\mathrm{Exp}[(\omega_m - b_g)\Delta t]_\times.}$$

Velocity:

$$\boxed{v_{k+1} = v_k + [R_k(a_m - b_a) + g]\Delta t.}$$

Position:

$$\boxed{p_{k+1} = p_k + v_k \Delta t + \frac{1}{2}[R_k(a_m - b_a) + g]\Delta t^2.}$$

ESKF error state:

$$\boxed{\delta x = [\delta p, \delta v, \delta\theta, \delta b_a, \delta b_g]^T.}$$

Error propagation:

$$\boxed{\delta \dot{x} = F\delta x + Gn}$$

and

$$\boxed{P_{k+1} = \Phi P_k \Phi^T + Q.}$$

Measurement residual:

$$\boxed{r = z - h(\hat{x}).}$$

Innovation covariance:

$$\boxed{S = HPH^T + R.}$$

Kalman gain:

$$\boxed{K = PH^T S^{-1}.}$$

Correction:

$$\boxed{\delta x = Kr.}$$

Orientation correction:

$$\boxed{R \leftarrow R\,\mathrm{Exp}([\delta\theta]_\times).}$$

Finally the estimator produces:

$$\boxed{T_{WE}(t) = \begin{bmatrix} R_{WE}(t) & p_{WE}(t) \\ 0 & 1 \end{bmatrix}}$$

together with velocity, biases and covariance.

For the camera/LiDAR system we've been building, this pose estimate is the **geometric backbone of the entire stack**: it lets you transform camera/LiDAR measurements into consistent frames, deskew individual LiDAR returns, accumulate multiple scans, estimate world-relative object velocity, compensate dynamic objects, and maintain persistent world-frame JPDA/MHT tracks.